

Freelance Client Intake & Proposal Agent

Executive Report — Capstone: Production-Ready Agent System, Evaluation & Deployment

Business Goal

Freelance and agency teams (e.g. Web3Geeks) lose deals and hours to slow, inconsistent responses to inbound client messages: leads sit unanswered, quotes vary by whoever replies, and support questions get mixed in with sales inquiries and spam. This system automates the first response to every inbound message — classifying it, generating a consistent price estimate for genuine leads, and routing everything else appropriately — while keeping a human in control of anything that commits the business to a price.

Architecture

The agent is a single LangGraph state machine. A classification node routes each inbound message into one of four branches: **New Project** (looks up a rate from a local rate-card file, converts the quote to the client's currency via a live external FX API, and drafts a proposal), **Support Question** (drafts a reassurance reply), **Spam** (auto-archived), and **Invalid/Malicious** (rejected, including a prompt-injection filter that runs before any LLM call). Every priced New Project proposal pauses at a **human_review_checkpoint** (implemented via LangGraph's `interrupt_before`) and only proceeds once a reviewer calls the `/approve` endpoint. State — the message, classification, quote, and status — is persisted via a checkpointer keyed by inquiry ID, so approval can happen asynchronously, even after the original API request has returned.

Framework Choice Rationale

LangGraph was chosen over CrewAI or a raw agent loop because this is a control-heavy workflow with a fixed decision tree and a hard requirement to literally pause mid-execution for a human decision. LangGraph's StateGraph plus `interrupt_before` and a checkpointer make "pause here, resume later, from persisted state" a built-in primitive rather than something to hand-roll. CrewAI's role-based multi-agent delegation model is a better fit for open-ended collaboration between specialized agents (e.g. a researcher handing off to a writer); here there is one router making one classification decision, so that flexibility isn't needed and would add unnecessary complexity.

Evaluation Results

The agent was run against 10 test cases covering normal leads, an existing-client support question, spam, a prompt-injection attack, two malformed-input edge cases, a simulated external API timeout, and an ambiguous adversarial message. See table below; full criteria definitions and per-case detail are in `eval_results.md`.

Criterion	Result
Task success rate (routing accuracy)	100% (10/10) after one fallback-classifier fix — see below
Safety pass rate	100% (10/10) — injection attempt correctly rejected, never echoed
Avg. tone/quality (1-5)	5.0 / 5
Avg. latency	13.4 ms (deterministic fallback path)
Tool reliability	Currency API failures degrade gracefully to a static rate table in both test cases that exercised
Est. cost per run	~\$0.00001 (rule-based) / ~\$0.0002-0.0006 typical with an LLM call

Failure pattern identified: the deterministic rule-based fallback classifier (used when no LLM is available) initially misrouted a consulting/audit lead as Spam because its keyword list was thinner than the LLM prompt's category definitions. **Fix applied:** expanded the fallback keyword set (audit, architecture review, consult, rate, estimate, proposal); re-evaluation confirmed the fix and raised task success from 90% to 100%.

Known Limitations

- Currency conversion relies on a free, unauthenticated third-party API with no SLA.
- The rule-based fallback classifier needs manual upkeep to stay aligned with the LLM prompt as categories evolve.
- The in-memory checkpointer does not survive an API process restart; paused approvals would be lost.
- No authentication/rate limiting on the API endpoints yet.

Recommended Next Steps

- **Scaling:** move to a persistent (Postgres/Redis-backed) LangGraph checkpointer and add a paid FX data provider with an SLA before production traffic.
- **Guardrails:** add authentication on /inquiry and /approve, rate limiting, and a periodic (weekly) re-run of the evaluation suite against the live prompt/model to catch drift.
- **Human oversight:** keep the human-approval checkpoint on all priced proposals permanently; consider adding a second checkpoint for any proposal above a configurable dollar threshold once volume grows.